আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম

الجامعة الإسلامية العالمية شيتاغونغ

International Islamic University Chittagong

Lab Report

COURSE CODE	: EEE- 4702
COURSE TITLE	: Control System I Sessional
EXPERIMENT NO	: 04
EXPERIMENT NAME	: Steady State error Analysis using MATLAB

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SESSION	: SPRING 2026
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REMARK:

9/10

Experiment No: 04

Experiment Name : Steady State Error Analysis using MATLAB

Objective :

1. To understand steady-state error of control systems
2. To identify system type (Type 0, Type 1, Type 2)
3. To observe step and ramp response using MATLAB
4. To analyze the effect of gain on system response
5. To determine steady-state error for different inputs

Apparatus:

1. Computer or Laptop
2. MATLAB software
3. Lab manual

Description of Task 1:

In this task, a system is given with forward path and feedback path. The transfer function depends on the student ID (ID = 96). The system is analyzed for different gain values .

Three types of systems are considered:

Type 0 system

Type 1 system (one integrator added)

Type 2 system (two integrators added)

For each system, step and ramp responses are obtained using MATLAB. The responses are plotted and compared to observe system behavior.

Description of Task 2:

In this task, a practical control system is analyzed. The gain k= k1+k2 is selected such that the system response is fast and the overshoot is less than 6%.

For ID = 6:

Maximum Overshoot:16%

Different values are tested. The best combination is selected based on rise time and overshoot. Then steady-state error for step and ramp input is calculated using MATLAB.

ET231096

MATLAB Representation of EXP:5

TASK :1

Editor - C:\Users\USER\Documents\MATLAB\EXP5_EEE4702_By_ET231096.m

EXP5_EEE4702_By_ET231096.m

```
clc;
clear;
close all;

K = [1 2 4 8 16];
ts0 = linspace(0,2,1001);
ts1 = 3*ts0;
ts2 = 20*ts0;

tr0 = 10*ts0;
tr1 = tr0;
tr2 = 2*tr0;
ID = 96;

nG = conv([1 ID],[1 ID+2]);
dG = conv([1 0 0],[1 4]);
dG = conv(dG,[1 ID+10]);
dG = conv(dG,[1 ID+15]);
nH = [1 1];
dH = conv([1 6],[1 10]);
n0 = nG;
d0 = dG;

n1 = n0;
d1 = [d0 0];      % Type 1
n2 = n1;
d2 = [d1 0];      % Type 2

for i = 1:length(K)
    for i1 = 1:3

        eval(['num = K(i)*n' num2str(i1-1) ';']);
        eval(['den = d' num2str(i1-1) ';']);
        G = tf(num,den);
        H = tf(nH,dH);
        T = feedback(G,H);
        eval(['ys' num2str(i1-1) '(:,' num2str(i) ') = step(T,ts' num2str(i1-1) ');']);
        t_temp = eval(['tr' num2str(i1-1)]);
        y_temp = lsim(T,t_temp,t_temp);
        eval(['yr' num2str(i1-1) '(:,' num2str(i) ') = y_temp;']);

    end
end

for i = 1:3

    eval(['ts = ts' num2str(i-1) ';']);
    eval(['ys = ys' num2str(i-1) ';']);
    eval(['tr = tr' num2str(i-1) ';']);
    eval(['yr = yr' num2str(i-1) ';']);
    figure(i); clf;
    set(gcf,'Position',[100 100 1000 500]);

    subplot(1,2,1)
    hold on
    for j = 1:length(K)
        plot(ts,ys(:,j),'LineWidth',1.5)
    end
    grid on
    xlabel('Time (s)')
    ylabel('Amplitude')
    title(['Type ' num2str(i-1) ' Step Response'])
    legend('K=1','K=2','K=4','K=8','K=16')
    hold off

    subplot(1,2,2)
    hold on
    for j = 1:length(K)
        plot(tr,yr(:,j),'LineWidth',1.5)
    end
    plot(tr,tr,'k--','LineWidth',1.5) % reference line
    grid on
    xlabel('Time (s)')
    ylabel('Amplitude')
    title(['Type ' num2str(i-1) ' Ramp Response'])
    legend('K=1','K=2','K=4','K=8','K=16','Reference')
    hold off
end
```

Fig 4.1 Steady State error Analysis using MATLAB Code task 1

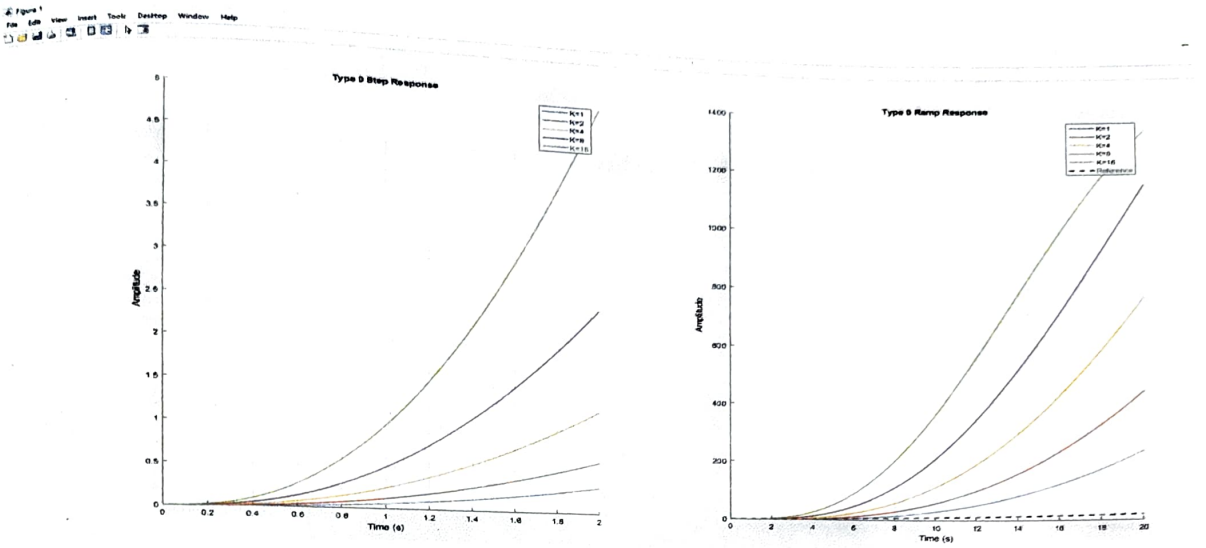

Fig 4.2 Step Response and Ramp Response Characteristics of Type 0

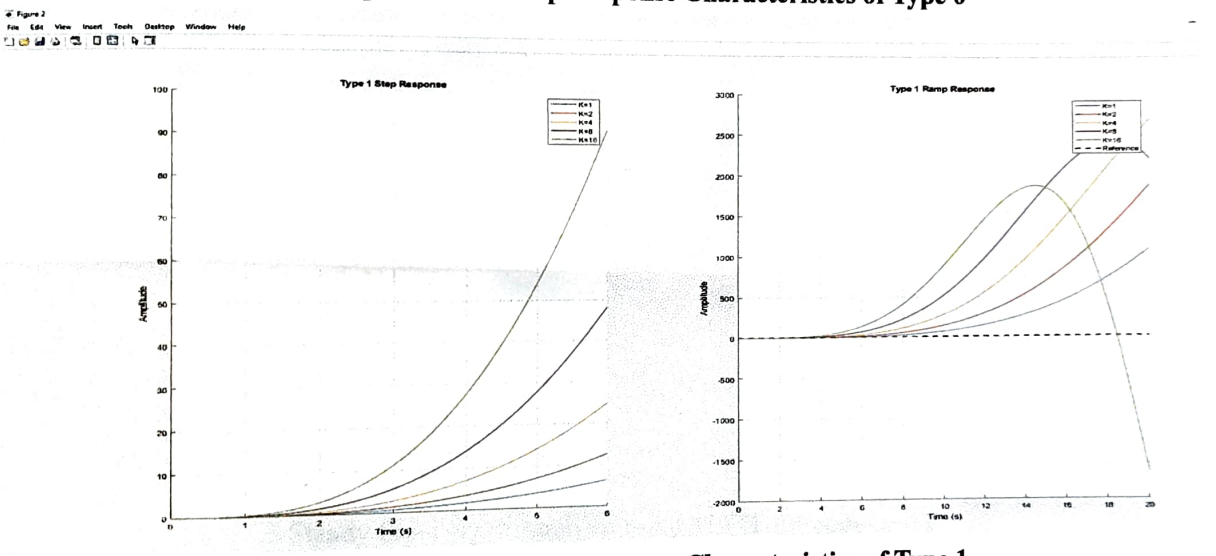

Fig 4.3 Step Response and Ramp Response Characteristics of Type 1

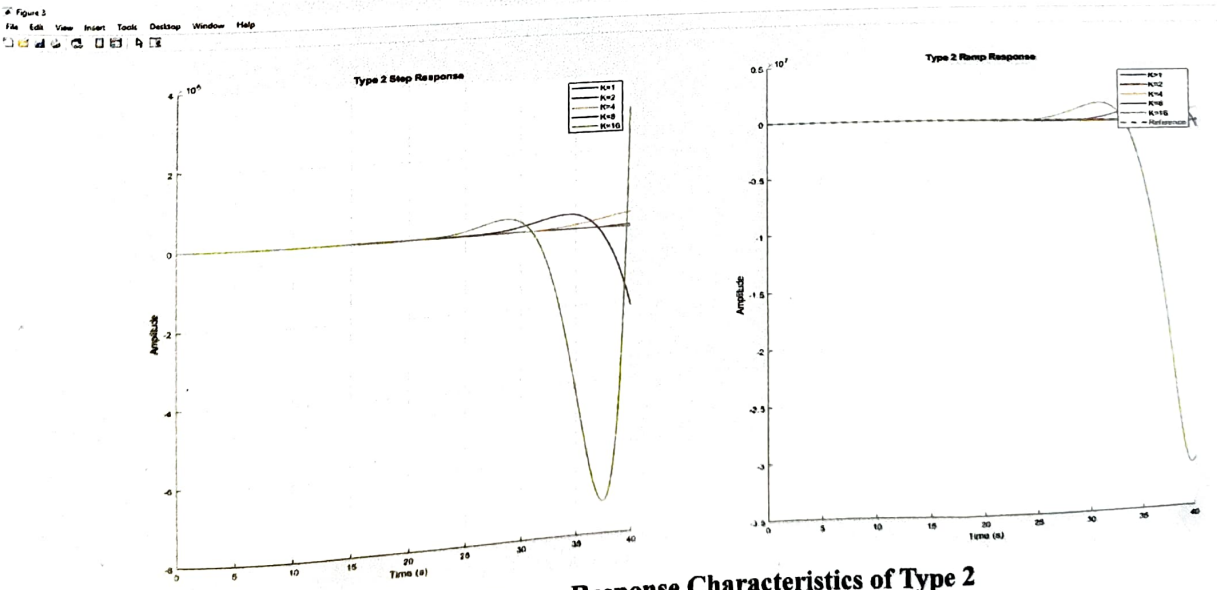

Fig 4.4 Step Response and Ramp Response Characteristics of Type 2

TASK 2

Editor - C:\Users\USER\Documents\MATLAB\EXP5_EEE4702_By_ET231096.m

EXP5_EEE4702_By_ET231096.m

```
clc;
clear;
close all;
%%My Student ID is ET231096
tau1=1;
tau2=0;
K1=4;
K2=4;

num=[tau1 1];
num1=K2*num;
den1=[tau2 1];

T1=tf(num1,den1);

num2=K1;
den2=[1 0 0];

T2=tf(num2,den2);

G=T1*T2;
T=feedback(G,1);

figure;
step(T);
title('Step Response By ET231096');
[y1,t1]=step(T);
figure;
plot(t1,1-y1);
title('Steady-State Error (Step) ET231096');
xlabel('Time'); ylabel('Error');
t=0:0.01:10;
ramp=t;
y2=lsim(T,ramp,t);

figure;
plot(t,ramp,'k--',t,y2,'b');
title('Ramp Response ET231096 ');
legend('Input','Output');

figure;
plot(t,ramp-y2);
title('Steady-State Error (Ramp) ET231096');
xlabel('Time'); ylabel('Error');
```

Fig 5.5 Steady State error Analysis using MATLAB code task 2

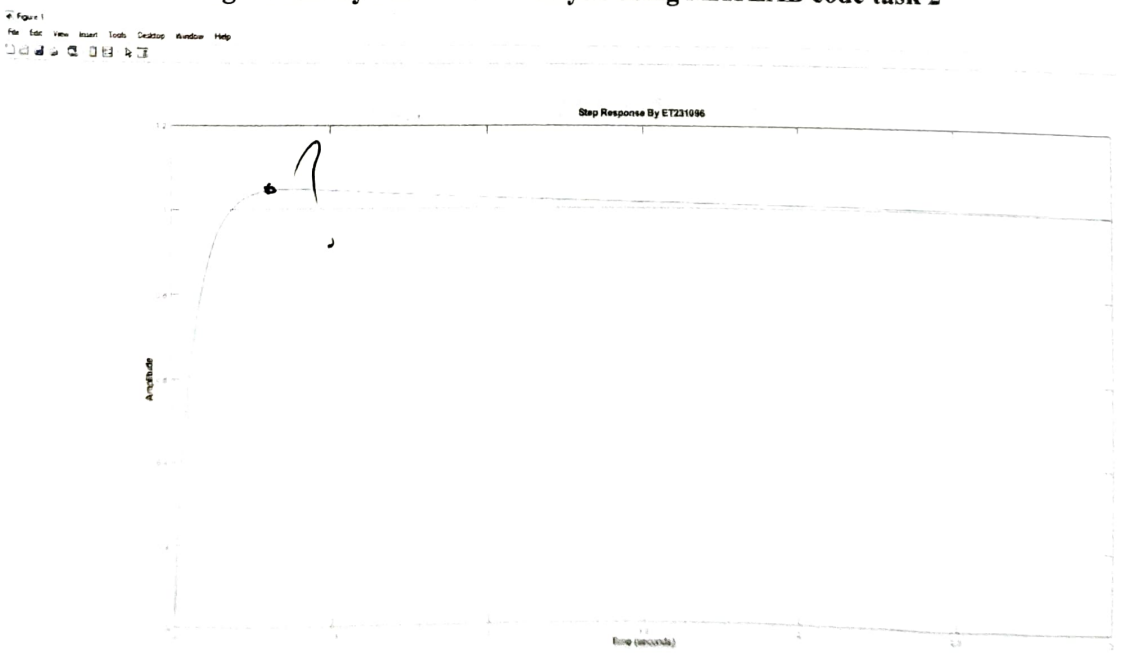

Fig 5.6 Step Response Characteristics

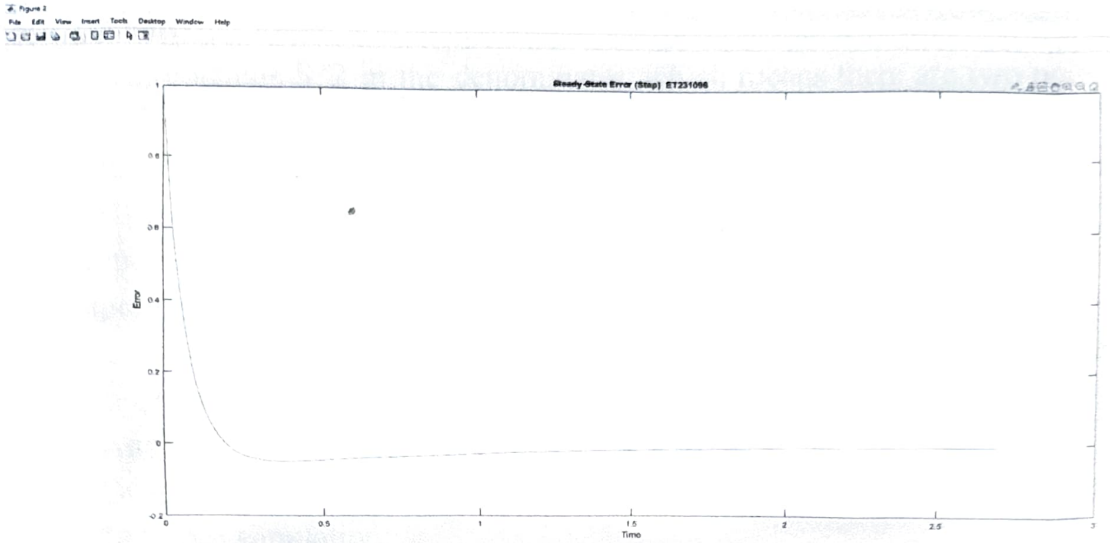

Fig 5.7 Steady state error Waveform

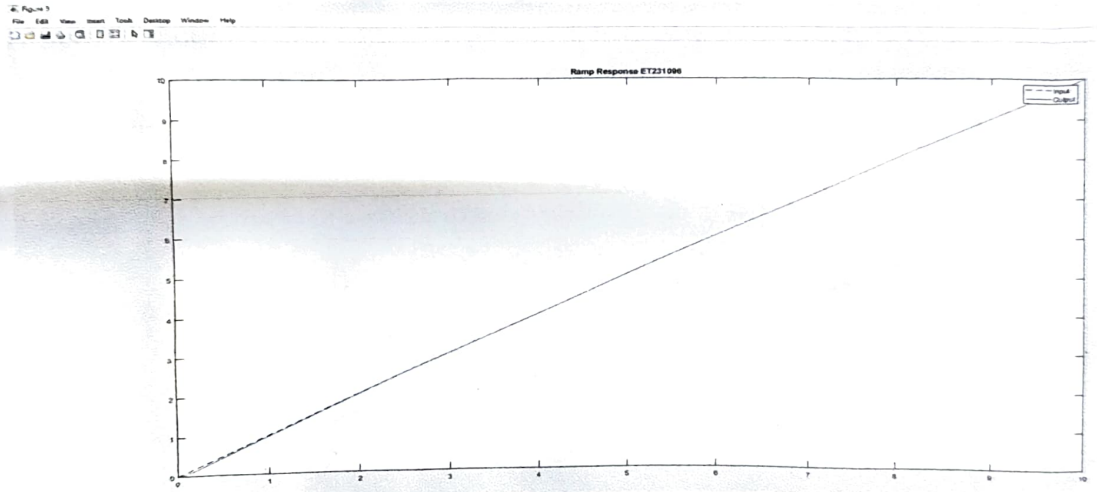

Fig 5.8 Ramp Response Waveform

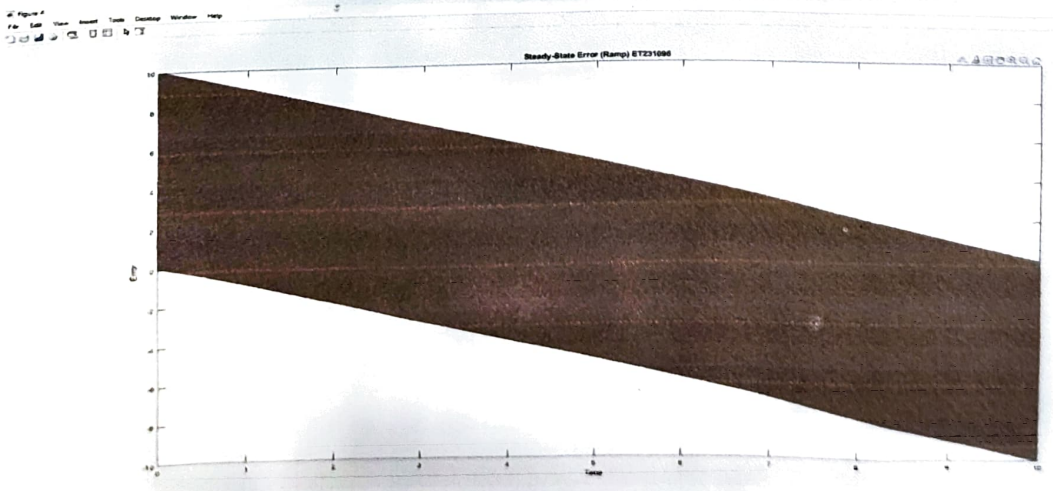

Fig 5.9 Steady state error (Ramp) Waveform

Result and Analysis:

Task 1 Analysis:

The system contains S^2 in the denominator, which means there are two poles at origin. Therefore, the system is a Type 2 system.

Step Input: For Type 2 system: e_ss = 0

MATLAB graph shows output reaches input value with no steady-state error.

Ramp Input: For Type 2 system: e_ss = 0

The output follows the ramp input closely. Error approaches zero.

Effect of Gain:Low gain = slow response, High gain= faster response but more oscillation, Gain improves tracking accuracy

Task 2 Analysis:

Given: Overshoot is less than 16%

From MATLAB simulation:

Best values: K1=4, k2= 4, K= 16

Transfer function:

(16s+16)/(S^2+16s+16)

The system shows fast response with acceptable overshoot and good stability.

Discussion:

In this experiment, steady-state error of different types of systems was studied using MATLAB. It was observed that type 2 systems provide zero steady state error for both step and ramp inputs. The effect of gain on system performance was also analyzed. Increasing gain improves response speed but may increase overshoot. From the practical system, a suitable gain value was selected that satisfies the given overshoot condition. The MATLAB result matched the theoretical analysis.